

Welcome

KIMQ8550 AIBOX USER MANUAL

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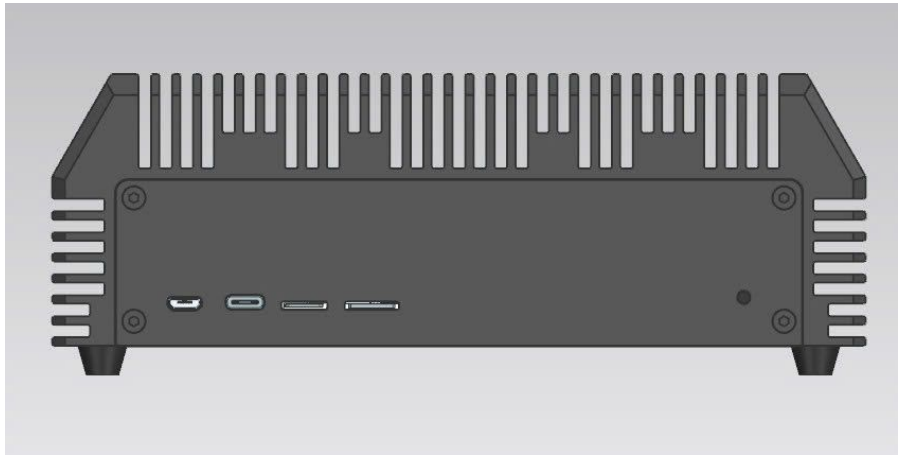
Preparation

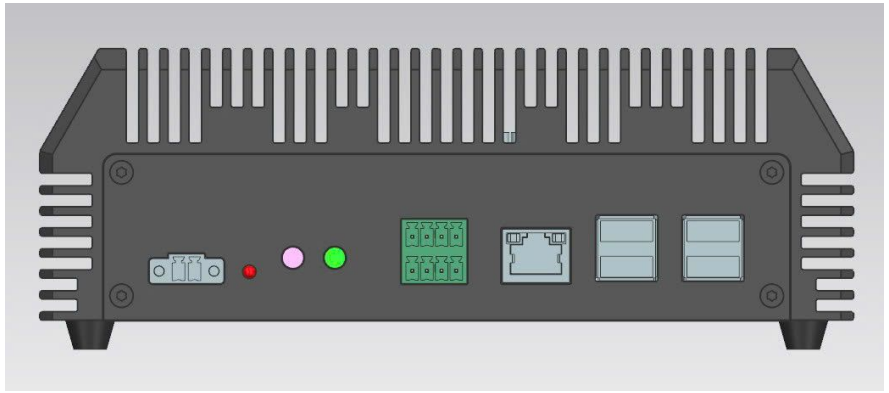
- **Ethernet cable for internet connect**
- **USB TYPE-C cable for flash OS**
- **Micro-usb cable to view log console via micro-usb connector**
- **Power adapter 12V DC**
- **Type-C to HDMI adapter to view on display via Type-C connector**
- **Ubuntu PC with adb, fastboot installed for flash images**

  sudo apt install minicom

  sudo apt install adb fastboot

  (https://developer.android.com/tools/releases/platform-tools)



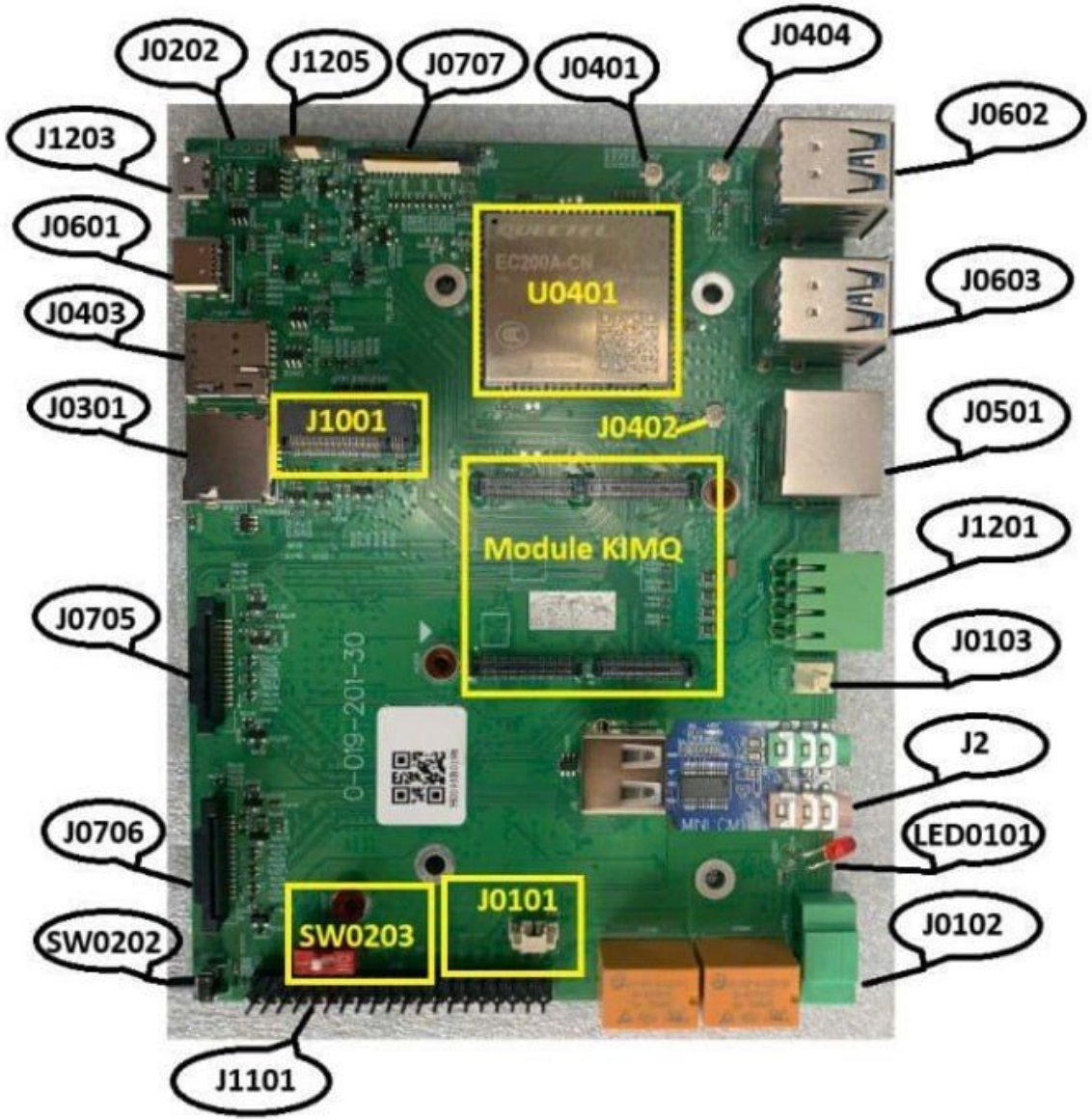


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Carrier Connection

Micro USB-(J1203)
USB Type-C 2.0 and Display port (J0601)
Nano Sim (J0403)
SD card (J0301)
SSD NVME (J1001)
Rear camera (J0705)
Front RGB Camera (J0706)
Reset Factory Button (SW0202)
Expansion Header (J1101)
Switch USB Mode (SW0203)
RTC Pin Connector (J0101)



Power Connector (J0102)
Power Led (LED 0101)
Audio Module (J2)
Fan Connector (J0103)
In/Out Connector (J1201)
Ethernet EJ45 (J0501)
USB Type-A 3.0 (J0602, J0603)
ANT_DRX Antenna (J0402)
GNSS Antenna (J0404)
Main Antenna (J0401)
Module EC200 (U0401)
LCD Connector (J1207)
Touch Connector (J1205)
Debug Header (J0202)

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Software Information

- **Ubuntu 22.04**
- **RAM : 16GB**
- **ROM : UFS 128 GB**
- **Use “apt update”, “apt-get install …” to install more packages if you need**
- **Use “dpkg -i --force-overwrite” to install deb file if apt-get update/upgrade failed to install**

  dpkg -i --force-overwrite /var/cache/apt/archives/libc6-arm64-cross_2.31-0ubuntu9.9cross1_all.deb

```
root@kalama:~# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.2 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.2 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-po"
UBUNTU_CODENAME=jammy
root@kalama:~#
```

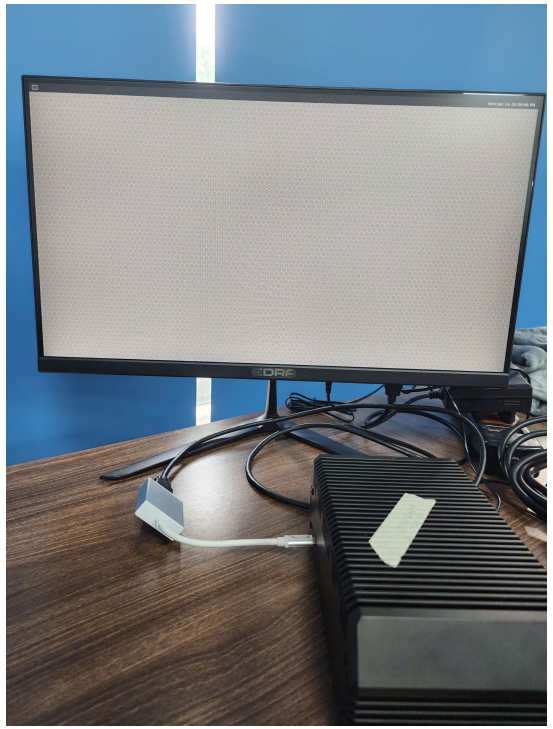
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Connect to Display via Type-C to HDMI adapter

- **Plugin power adapter to Box**
- **Use “Type-C to HDMI” adapter to connect Box with Display via DisplayPort interface**
- **After HDMI display work, you can plug in mouse and keyboard to 4 ports TYPE-A to work with BOX**





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Access to Carrier Board

- **Plug ethernet cable into RJ45 connector**
- **On Ubuntu HOST PC, use micro-usb cable connect to AIBOX then open /dev/ttyACM0 to view console boot log**

  sudo minicom -D /dev/ttyACM0 -b 115200
- **Plug in power adapter and AIBOX will boot and be allocated DHCP IP address,**
- **On terminal console log, type "root/oelinux123" to login into AIBOX and check IP by command "ifconfig eth0"**

- **By adb command via USB Type-C cable**

  $ adb devices

  $ adb shell
- **By ssh command**

  $ ssh root@<IP_address> -> Type password "oelinux123"

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Mount folder between Box and Host PC

For easy access and convenience in editing and updating data, the box and computer can be connected in the following two ways:

- **Mount folder of Box from Host PC**

  $ sudo apt install sshfs

  $ mkdir ~/mount

  $ sshfs root@192.168.1.189:/data/ ~/mount -> Then /data partition of box will be visible at ~/mount of Host PC

- **Mount folder of Host PC from Box**

  # apt install sshfs

  # mkdir /home/mount

  # sshfs hsptek@192.168.1.25:~/test-data/ /home/mount

- **To umount mounted directory**

  fusermount -u ~/mount

  or

  fusermount -u /home/mount

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GPIO control

- View system GPIO configuration status

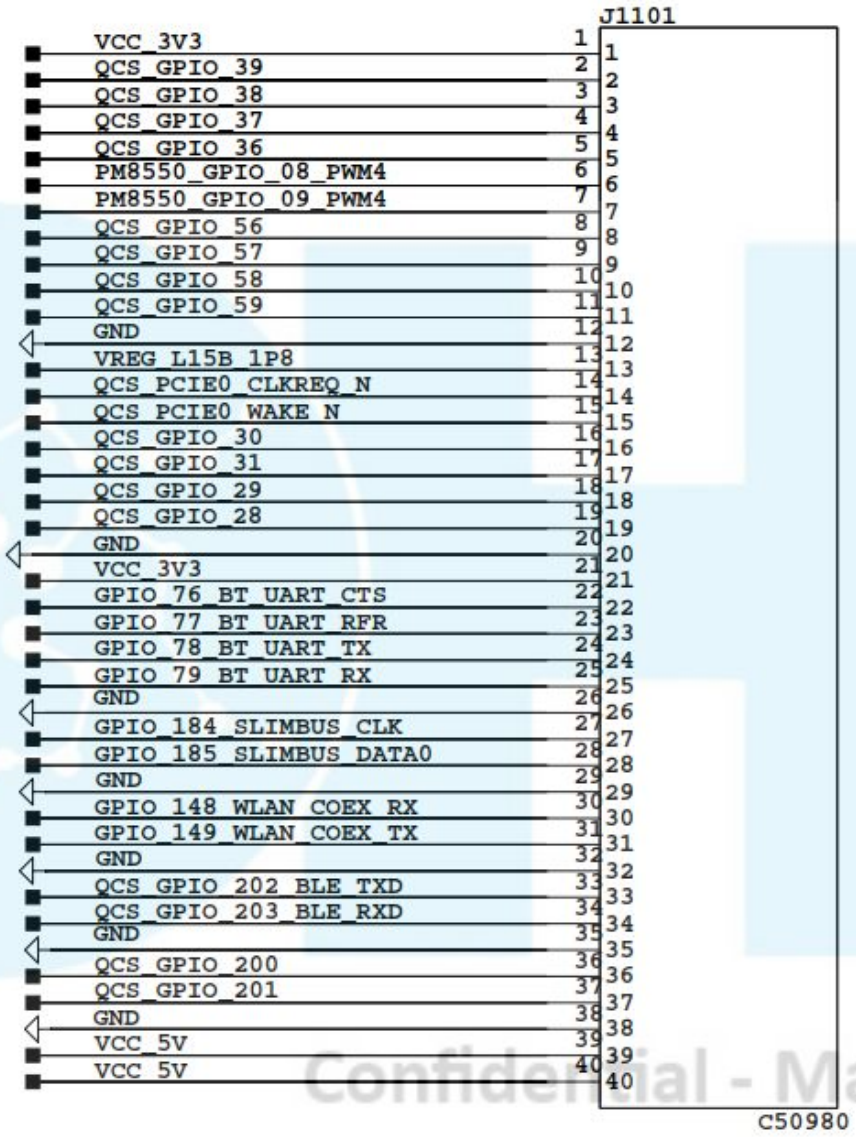
  cat /sys/kernel/debug/gpio

```
gpiochip0: GPIOs 301-511, parent: platform/f000000.pinctrl, f000000.pinctrl:
 gpio0   : in  low  func0 2mA pull down
 gpio1   : in  low  func0 2mA pull down
 gpio2   : in  low  func0 2mA pull down
 gpio3   : in  low  func0 2mA pull down
 gpio4   : in  high func0 2mA pull up
 gpio5   : in  high func0 2mA pull up
 gpio6   : out high func0 2mA pull down
 gpio7   : in  low  func0 2mA pull down
 gpio8   : in  low  func0 2mA pull down
 gpio9   : in  low  func0 2mA pull down
 gpio10  : in  low  func0 2mA pull down
```

- Control GPIO in user space

To export GPIO, GPIO number need to be determined by checking "**cat /sys/kernel/debug/gpio**" instead of using the hardware numbering.
For example, to export and control PIN number 2 of J1101 (GPIO_39 of QCS8550), refer to GPIO configuration status, we need to add 301 to the hardware numbering, we got number 340 to export

```
# cd /sys/class/gpio
//export gpio_39
# echo 340 > export
//Configure as an output mode pin
# cd gpio340
# echo out > direction
//Configure to output a high-level voltage
# echo 1 > value
//Configure to output a low-level voltage
# echo 0 > value
//Configure as an input mode pin
#echo in > direction
```



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Flash OS images

- **Enter fastboot mode**

  By adb via Type-C cable

  $ adb devices

  $ adb root

  $ adb reboot bootloader

  Or you can enter fastboot mode via micro-usb cable console by command:
  systemctl reboot bootloader

- **Flash images**

  $ fastboot flash abl_a abl.elf

  $ fastboot flash abl_b abl.elf

  $ fastboot flash dtbo_a dtbo.img

  $ fastboot flash dtbo_b dtbo.img

  $ fastboot flash boot_a boot.img

  $ fastboot flash boot_b boot.img

  $ fastboot flash system system.img

  $ fastboot reboot

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Qualcomm AI Runtime (QAIRT)

1. Follow steps and create Qualcomm account using your email at https://myaccount.qualcomm.com/signup (ignore this step if you've already have one)
2. Take a look about QAIRT at link https://docs.qualcomm.com/doc/80-63442-10/topic/general_overview.html
3. Login and download lastest **QAIRT** version by click "**Get Software**" at bellow links:
   SNPE:
   https://qpm.qualcomm.com/main/tools/details/qualcomm_neural_processing_sdk
   QNN:
   https://www.qualcomm.com/developer/software/qualcomm-ai-engine-direct-sdk

   (Both 2 links download same QAIRT (QNN & SNPE) SDK)

1. Extract downloaded zip file and follow guide at ${SNPE_ROOT}/docs/QAIRT-Docs/index.html or above QAIRT overview link

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DSP Runtime support on Box

On Host PC with SNPE downloaded:

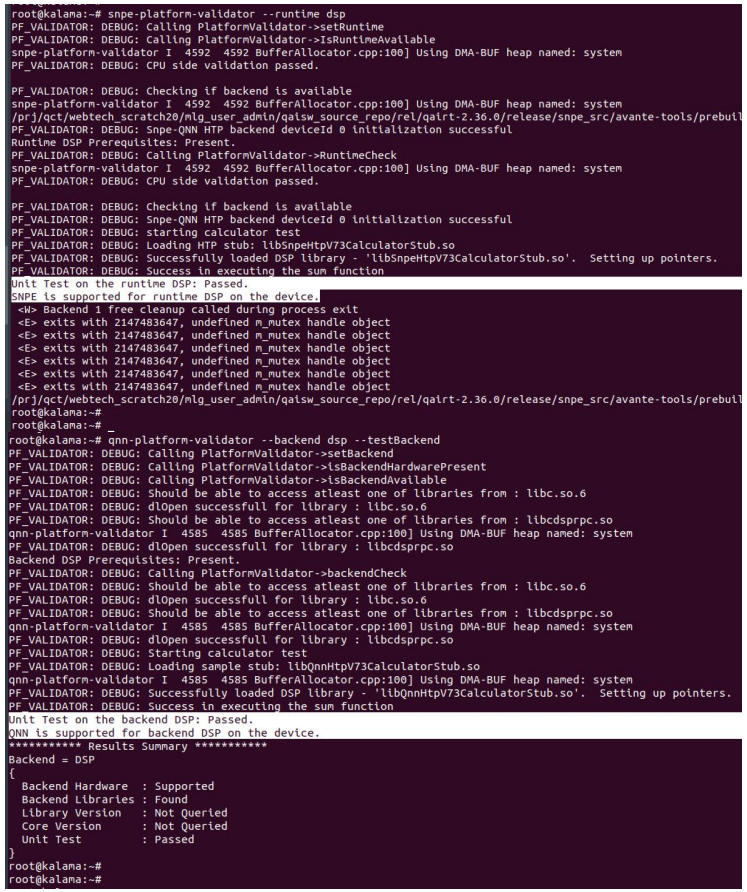
/////// PREPARE (do one time only) ///////

/////// push SNPE libs into the device ///////

```
// IP of the BOX. change to yours
export IP=192.168.1.100
// SNPE SDK path. change to yours
export SNPE_ROOT=/media/hsptek/data/aistack/2.41.0.251128
export ARM_ARCH=aarch64-oe-linux-gcc11.2
scp $SNPE_ROOT/lib/$ARM_ARCH/* root@$IP:/usr/lib/
scp $SNPE_ROOT/lib/hexagon-v73/unsigned/* root@$IP:/usr/lib/rfsa/adsp/
scp $SNPE_ROOT/bin/$ARM_ARCH/* root@$IP:/usr/bin/
```

Reboot device and access into box, check command:

```
snpe-platform-validator --runtime dsp
qnn-platform-validator --backend dsp --testBackend
```



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Run example yolonas demo

Preparation:

KIMQ8550 AIBOX with display connect via Type-C to HDMI adapter

Download yolodemo.zip file, extract and push into device AI

export IP=192.168.1.100 # change to yours

export YOLO=~/yolodemo # change to yours

scp $YOLO/yolo_nas_s_v213_quantized.dlc $YOLO/yolov5m.labels root@$IP:/data/

Run example:

With video test file

export XDG_RUNTIME_DIR=/run/user/root

export WAYLAND_DISPLAY=wayland-1

gst-launch-1.0 qtivcomposer name=mixer sink_1::dimensions="<1920,1080>" ! queue ! waylandsink sync=true fullscreen=true \

filesrc location=/data/video.mp4 ! qtdemux ! queue ! h264parse ! qtic2vdec ! queue ! tee name=t ! queue ! mixer. t. ! queue ! \

qtimlvconverter ! queue ! \

qtimlsnpe delegate=dsp model=/data/yolo_nas_s_v213_quantized.dlc layers="< /heads/Mul, /heads/Sigmoid >" ! queue ! \

qtimlvdetection threshold=51.0 results=10 module=yolo-nas labels=/data/yolov5m.labels ! \

video/x-raw,width=480,height=270 ! queue ! mixer.

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Run example yolonas demo

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Download yolodemo.zip file, extract and push into device AI

export IP=192.168.1.100 # change to yours

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scp $YOLO/yolo_nas_s_v213_quantized.dlc $YOLO/yolov5m.labels root@$IP:/data/

Run example:

With RTSP H265 stream

export XDG_RUNTIME_DIR=/run/user/root

export WAYLAND_DISPLAY=wayland-1

export RTSP='rtsp://admin:Hsptek2024@192.168.1.200:554/Streaming/Channels/101'

gst-launch-1.0 qtivcomposer name=mixer sink_1::dimensions="<1920,1080>" ! queue ! waylandsink sync=true fullscreen=true \

rtspsrc location=$RTSP ! rtph265depay ! queue ! h265parse ! qtic2vdec ! queue ! tee name=t ! queue ! mixer. t. ! queue ! \

qtimlvconverter ! queue ! \

qtimlsnpe delegate=dsp model=/data/yolo_nas_s_v213_quantized.dlc layers="< /heads/Mul, /heads/Sigmoid >" ! queue ! \

qtimlvdetection threshold=51.0 results=10 module=yolo-nas labels=/data/yolov5m.labels ! \

video/x-raw,width=480,height=270 ! queue ! mixer.

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Run example yolonas demo

Preparation:

KIMQ8550 AIBOX with display connect via Type-C to HDMI adapter

Download yolodemo.zip file, extract and push into device AI

export IP=192.168.1.100 # change to yours

export YOLO=~/yolodemo # change to yours

scp $YOLO/yolo_nas_s_v213_quantized.dlc $YOLO/yolov5m.labels root@$IP:/data/

Run example:

With RTSP H264 stream

export XDG_RUNTIME_DIR=/run/user/root

export WAYLAND_DISPLAY=wayland-1

export RTSP='rtsp://admin:Hsptek2024@192.168.1.200:554/Streaming/Channels/101'

gst-launch-1.0 qtivcomposer name=mixer sink_1::dimensions="<1920,1080>" ! queue ! waylandsink sync=true fullscreen=true \

rtspsrc location=$RTSP ! rtph264depay ! queue ! h264parse ! qtic2vdec ! queue ! tee name=t ! queue ! mixer. t. ! queue ! \

qtimlvconverter ! queue ! \

qtimlsnpe delegate=dsp model=/data/yolo_nas_s_v213_quantized.dlc layers="< /heads/Mul, /heads/Sigmoid >" ! queue ! \

qtimlvdetection threshold=51.0 results=10 module=yolo-nas labels=/data/yolov5m.labels ! \

video/x-raw,width=480,height=270 ! queue ! mixer.

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Set timezone

Use command timedatectl

For example: To set timezone GMT+7:

timedatectl set-timezone Asia/Ho_Chi_Minh

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Install docker on box

1. Access into box via adb or ssh, run below command to install docker

```
# apt update

# apt-get install apt-transport-https ca-certificates curl gnupg2 software-properties-common
# curl -fsSL https://download.docker.com/linux/ubuntu/gpg | apt-key add -
# apt-key fingerprint 0EBFCD88
# add-apt-repository "deb [arch=arm64] https://download.docker.com/linux/ubuntu bionic stable"
# apt-get update
# apt-get install docker-ce docker-ce-cli containerd.io
```

1. Test docker

```
docker run hello-world
```

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Run docker on box

After install docker successful, please follow below steps to run docker on box:

////////// Pull and run docker image //////////

docker pull arm64v8/ubuntu:22.04

// check image to see the IMAGE ID

docker images

// run the docker, replace fde9c12d7d3f to your own IMAGE ID

docker run -it --privileged -v /data:/data -v /usr:/usr -v /lib:/lib -v /dsp:/dsp -v /system:/system -v /vendor:/vendor -v /dev:/dev -v /etc:/etc -v /lib64:/lib64 -v /mnt:/mnt -v /opt:/opt -v /proc:/proc -v /run:/run -v /sbin:/sbin -v /sys:/sys -v /target:/target -v /var:/var --device=/dev/ 3048ba078595 bash

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Set static IP

Open file /etc/dhcpcd.conf and add bellow lines at the end of file (change IP and gateway to yours)

interface eth0

static ip_address=192.168.1.229/24

static routers=192.168.1.1

static domain_name_servers=192.168.1.1

```
# Generate SLAAC address using the Hardware Address of the interface
#slaac hwaddr
# OR generate Stable Private IPv6 Addresses based from the DUID
slaac private

interface eth0
static ip_address=192.168.1.229
static routers=192.168.1.1
static domain_name_servers=192.168.1.1
root@qcs6490-odk:~#
```

```
root@qcs6490-odk:~# ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.229  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 2405:4800:1f1d:f000:9389:5403:4cb3:8acc  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::1d8e:1afb:624a:44a4  prefixlen 64  scopeid 0x20<link>
        inet6 2405:4800:1f1d:f000:21a6:a259:1241:edab  prefixlen 64  scopeid 0x0<global>
        ether d8:80:39:11:22:42  txqueuelen 1000  (Ethernet)
        RX packets 15369  bytes 1253995 (1.2 MB)
        RX errors 0  dropped 1  overruns 0  frame 0
        TX packets 914  bytes 213779 (213.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@qcs6490-odk:~#
```

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